AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI

ANNUAL EXAMINATIONS (THEORY) 2024

Chemistry Paper II

Time: 1 hour 30 minutes Marks: 35

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INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's Signature**

RUBRIC

2. There are EIGHT questions. Answer ALL questions. Questions 7 & 8 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue correcting fluid, or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a scientific calculator if you wish.

Q.1. (Total 4 Marks)

a. What will be observed if a small paddle wheel is placed in the path of cathode rays? Write the property of cathode rays that supports your answer. (2 Marks)

_____

_____

_____

_____

b. Write the electronic configuration of an element **X** with atomic number 15 using the following rules:

i. Aufbau principle (1 Mark)

_____

_____

ii. Hund's rule (1 Mark)

_____

_____

Q.2. (Total 2 Marks)

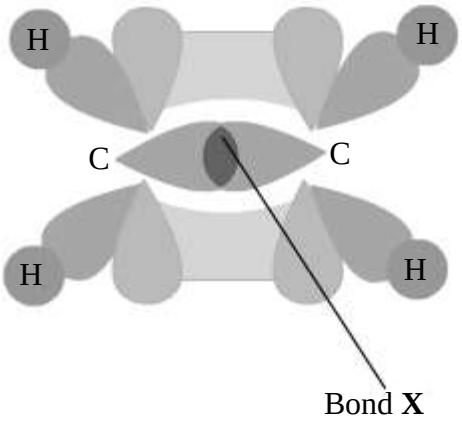
Compare the given attributes of a bond in the following gas molecules.

Attribute	**O_2**	**N_2**
Bond Energy		
Bond Strength		

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Q.3. (Total 3 Marks)

Consider the orbital shape of a molecule.



Identify the following in the given diagram.

a. Hybridisation: _____

b. Geometry: _____

c. Bond **X**: _____

Q.4. (Total 4 Marks)

In a closed container at 900 K, 6.0×10^{-3} M of $CH_{4(g)}$ reacts with 4.0×10^{-3} M of $H_2O_{(g)}$ to produce 8.0×10^{-3} M of $CO_{(g)}$ and 3.0×10^{-3} M of $H_{2(g)}$. The equation for the reversible reaction is as under.

$CH_{4(g)} + H_2O_{(g)} \rightleftharpoons CO_{(g)} + 3H_{2(g)}$ $K_c = 2.4 \times 10^{-4}$

a. Calculate the value of Q_c. (2 Marks)

_____

_____

_____

_____

b. Determine the direction in which the reaction will proceed. (2 Marks)

_____

_____

_____

_____

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Q.5. (Total 4 Marks)

a. **X** and **Y** react to form a product **Z**. In the first experiment (first run), the concentration of **X** is doubled while **Y** is held constant, and the initial rate doubles. In the second experiment (second run), the concentration of **Y** is doubled while the concentration of **X** remains constant, and the initial rate quadruples.

Deduce the order of the reaction with respect to **X** and **Y**. (2 Marks)

(**Note**: Assume that the term 'constant' represents a concentration of 1 mol dm^{-3}.)

b. Consider the following results for the decomposition of dinitrogen pentaoxide.

$$2N_2O_{5(g)} \rightarrow 4NO_{2(g)} + O_{2(g)}$$

Concentration of N_2O_5 (mol dm^{-3})	**Initial Rate (mol dm^{-3} s^{-1})**
0.0016	0.12
0.0024	0.18
0.0032	0.24

Calculate the initial rate of reaction when the concentration of N_2O_5 is 0.002 mol dm^{-3}. (2 Marks)

(**Note**: Show the complete steps of working.)

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Q.6. (Total 4 Marks)

Consider a 16 g sample of methanol (CH_3OH) is burnt in the presence of excess oxygen in a bomb calorimeter containing 4,000 g of water. The temperature of the water increases from 22°C to 27°C.

If the heat capacity of the calorimeter is 2,657 J/°C and the specific heat of water is 4.2 J/g°C, then calculate the enthalpy of combustion of methanol in kJ mol^{-1}.

(**Note**: Molar mass of methanol is 32 g mol^{-1}.)

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Q.7. (Total 7 Marks)

EITHER

a. According to the following equation, 96 g of sulphur reacts with 36 g of carbon to produce carbon disulphide.

$$2S_{(s)} + C_{(s)} \xrightarrow{\text{heat}} CS_{2(l)}$$

Calculate the following and identify the limiting reagent in this reaction.

i. The mass of carbon disulphide produced (5 Marks)

ii. The excess amount of the reactant in grams that is left un-reacted (2 Marks)

(**Note**:

- ${}^{12}_{6}C$ and ${}^{32}_{16}S$
- Show the complete steps of working.)

OR

b. Balance the following reaction using the oxidation number method.

$MnO_4^{1-} + Fe^{2+} + H^{1+} \rightarrow Mn^{2+} + Fe^{3+} + H_2O$ (7 Marks)

(**Note**: Show the complete steps of working.)

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Q.8. (Total 7 Marks)

EITHER

a. When silver chloride dissolves in a solution of ammonia, it results in the formation of a complex ion, $Ag(NH_3)_2^+$.

Explain $Ag(NH_3)_2^+$ as a Lewis acid-base adduct by including the following in your answer.

i. Classification of reactants along with the justification for each classification (4 Marks)

ii. Balanced equation for the dissolution of AgCl in a solution of NH_3 (1 Mark)

iii. Lewis structure of $Ag(NH_3)_2^+$ with the identification of the type of bond present in it (2 Marks)

OR

b. Compare true solutions with suspensions based on the given attributes. (7 Marks)

(**Note**: Write the points of comparison in tabular form.)

i. Nature of the solutions

ii. External appearance

iii. Size of the particles (in diameter)

iv. Diffusion of the solution through parchment and filter paper

v. Sedimentation

vi. Visibility of particles

vii. Brownian movements

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